

Regulation of caspases in the nervous system implications for functions in health and disease.

PMID: 21238939 | Indexed in PubMed

Caspases, initially identified as a family of proteases regulating cell death, have been found to have nonapoptotic functions as well. Some family members are critical for mediating programmed cell death in development. After development, caspases are downregulated in the nervous system, but continue to perform important nonapoptotic functions relevant for neurogenesis and synaptic plasticity. In neurodegenerative diseases, where aberrant neuronal death is an outstanding feature, there is an increase in caspase activity. The specific caspase death pathways leading to dysfunction and death have still not been fully clarified, despite the plethora of scientific literature addressing these issues. In this chapter, we will present the current knowledge of caspase activation and activity pathways, the current tools for examining caspases, and functions of caspases in the nervous system in health and in disease. Alzheimer's Disease, the most common neurodegenerative disorder, and cerebral ischemia, the most common cause of neurologic death, are used to illustrate our current understanding of death signaling in neurodegenerative diseases. A better understanding of how caspases function in health and disease would provide appropriate specific targets for the development of therapeutic interventions for these diseases. Life and death are exquisitely regulated at the cellular level from development through maturity. During development, neuronal death is the major factor shaping the nervous system. This death is mainly caspase-mediated apoptosis. Once the waves of developmental death have passed (death occurs at different times in different parts of the nervous system), there is downregulation of the death machinery, as the postmitotic neurons should live for the life of the organism. Aberrant neuronal death is a major part of neurodegenerative disorders, but there is still no clear understanding of the processes leading to the phenotypes of the various diseases. Even the type of death that occurs continues to be debated, whether it is apoptotic, necrotic, or autophagic, or some combination of these death mechanisms. Here, we will discuss the role that the caspases play in neuronal function, dysfunction, and death. First, we will discuss the regulation of caspase activation and activity. We will examine the current understanding of caspase function in developmental neuronal death and then illustrate the role of caspases in neuronal death in disease employing two diseases of neuronal loss, Alzheimer's Disease (AD), which is the most common chronic neurodegenerative disorder, and cerebral ischemia/stroke, the third most common cause of death in Western society, which is an acute neuronal disorder with chronic sequelae.

Induction of heat shock proteins in differentiated human neuronal cells following co-application of celastrol and arimoclomol.

PMID: 27273088 | Indexed in PubMed

Few effective therapies exist for the treatment of neurodegenerative diseases that have been characterized as protein misfolding disorders. Upregulation of heat shock proteins (Hsps) mitigates against the accumulation of misfolded, aggregation-prone proteins and synaptic dysfunction, which is recognized as an early event in neurodegenerative diseases. Enhanced induction of a set of Hsps in differentiated human SH-SY5Y neuronal cells was observed following co-application of celastrol and arimoclomol, compared to their individual application. The dosages employed did not affect cell viability or neuronal process morphology. The induced Hsps included the little studied HSPA6 (Hsp70B'), a potentially neuroprotective protein that is present in the human genome but not in rat and mouse and

hence is missing in current animal models of neurodegenerative disease. Enhanced induction of HSPA1A (Hsp70-1), DNAJB1 (Hsp40), HO-1 (Hsp32), and HSPB1 (Hsp27) was also observed. Celastrol activates heat shock transcription factor 1 (HSF1), the master regulator of Hsp gene transcription, and also exhibits potent anti-inflammatory and anti-oxidant activities. Arimoclomol is a co-activator that prolongs the binding of activated HSF1 to heat shock elements (HSEs) in the promoter regions of inducible Hsp genes. Elevated Hsp levels peaked at 10 to 12 h for HSPA6, HSPA1A, DNAJB1, and HO-1 and at 24 h for HSPB1. Co-application of celastrol and arimoclomol induced higher Hsp levels compared to heat shock paired with arimoclomol. The co-application strategy of celastrol and arimoclomol targets multiple neurodegenerative disease-associated pathologies including protein misfolding and protein aggregation, inflammatory and oxidative stress, and synaptic dysfunction.

Neuropathological spectrum of synucleinopathies.

PMID: 14502650 | Indexed in PubMed

Synucleinopathies comprise a diverse group of neurodegenerative proteinopathies that share common pathological lesions composed of aggregates of conformational and posttranslational modifications of alpha-synuclein in selected populations of neurons and glia. Abnormal filamentous aggregates of misfolded alpha-synuclein protein are the major components of Lewy bodies, dystrophic (Lewy) neurites, and the Papp-Lantos filaments in oligodendroglia and neurons in multiple system atrophy linked to degeneration of affected brain regions. The synucleinopathies include (1) Lewy body disorders and dementia with Lewy bodies, (2) multiple system atrophy (MSA), and (3) Hallervorden-Spatz disease. (1) The pathological diagnosis of Lewy body disorders and dementia with Lewy bodies is established by validated consensus criteria based on semiquantitative assessment of subcortical and cortical Lewy bodies as their common hallmarks. They are accompanied by subcortical multisystem degeneration with neuronal loss and gliosis with or without Alzheimer pathologic state. Lewy bodies also occur in numerous other disorders, including pure autonomic failure, neuroaxonal dystrophies, and various amyloidoses and tauopathies. (2) Multiple system atrophy, a sporadic, adult-onset degenerative movement disorder of unknown cause, is characterized by alpha-synuclein-positive glial cytoplasmic and rare neuronal inclusions throughout the central nervous system associated with striatonigral degeneration, olivopontocerebellar atrophy, and involvement of medullar and spinal autonomic nuclei. (3) In neurodegeneration with brain iron accumulation type I, or Hallervorden-Spatz disease, alpha-synuclein is present in axonal spheroids and glial and neuronal inclusions. While the identity of the major components of Lewy bodies suggests that a pathway leading from normal soluble to abnormal misfolded filamentous proteins is central for their pathogenesis, regardless of the primary disorder, there are conformational differences in alpha-synuclein between neuronal and glial aggregates, showing nonuniform mapping for its epitopes. Despite several cellular and transgenic models, it is not clear whether inclusion body formation is an adaptive/neuroprotective or a pathogenic reaction/process generated in response to different, mostly undetermined, functional triggers linked to neurodegeneration.

Prions: generation and spread versus neurotoxicity.

PMID: 24860100 | Indexed in PubMed

Neurodegenerative diseases are characterized by the aggregation of misfolded proteins in the brain. Among these disorders are the prion diseases, which are transmissible, and in which the misfolded

proteins ("prions") are also the infectious agent. Increasingly, it appears that misfolded proteins in Alzheimer and Parkinson diseases and the tauopathies also propagate in a "prion-like" manner. However, the association between prion formation, spread, and neurotoxicity is not clear. Recently, we showed that in prion disease, protein misfolding leads to neurodegeneration through dysregulation of generic proteostatic mechanisms, specifically, the unfolded protein response. Genetic and pharmacological manipulation of the unfolded protein response was neuroprotective despite continuing prion replication, hence dissociating this from neurotoxicity. The data have clear implications for treatment across the spectrum of these disorders, targeting pathogenic processes downstream of protein misfolding.

TDP-43 and FUS en route from the nucleus to the cytoplasm.

PMID: 28380257 | Indexed in PubMed

Misfolded or mislocalized RNA-binding proteins (RBPs) and, consequently, altered mRNA processing, can cause neuronal dysfunction, eventually leading to neurodegeneration. Two prominent examples are the RBPs TAR DNA-binding protein of 43 kDa (TDP-43) and fused in sarcoma (FUS), which form pathological messenger ribonucleoprotein aggregates in patients suffering from amyotrophic lateral sclerosis (ALS) and frontotemporal dementia (FTD), two devastating neurodegenerative disorders. Here, we review the multiple functions of TDP-43 and FUS in mRNA processing, both in the nucleus and in the cytoplasm. We discuss how TDP-43 and FUS may exit the nucleus and how defects in both nuclear and cytosolic mRNA processing events, and possibly nuclear export defects, may contribute to neurodegeneration and ALS/FTD pathogenesis.

Rescuing neurons in prion disease.

PMID: 21158739 | Indexed in PubMed

One of the major current challenges to both medicine and neuroscience is the treatment of neurodegenerative diseases, which pose an ever-increasing medical, social and economic burden in the developed world. These disorders, which include Alzheimer's, Huntington's and Parkinson's diseases, and the rarer prion diseases, are separate entities clinically but have common features, including aggregates of misfolded proteins and varying patterns of neurodegeneration. A key barrier to effective treatment is that patients present clinically with advanced, irreversible, neuronal loss. Critically, mechanisms of neurotoxicity are poorly understood. Prevention of neuronal loss, ideally by targeting underlying pathogenic mechanisms, must be the aim of therapy. The present review describes the rationale and experimental approaches that have allowed such prevention, rescuing neurons in mice with prion disease. This rescue cured animals of a rapidly fatal neurodegenerative condition, resulting in symptom-free survival for their natural lifespan. Early pathological changes were reversed; behavioural, cognitive and neurophysiological deficits were recovered; and there was no neuronal loss. This was achieved by targeting the central pathogenic process in prion disease rather than the presumed toxic species, first by proof-of-principle experiments in transgenic mice and then by treatment using RNA interference for gene knockdown. The results have been a new therapeutic target for prion disease, further insight into mechanisms of prion neurotoxicity and the discovery of a window of reversibility in neuronal damage. Furthermore, the work gives rise to new concepts for treatment strategies for other neurodegenerative disorders, and highlights the need for clinical detection of early neuronal dysfunction, so that similar early rescue can also be achieved for these disorders.

The role of activity in synaptic degeneration in a protein misfolding disease, prion disease.

PMID: 22815961 | Indexed in PubMed

In chronic neurodegenerative diseases associated with aggregates of misfolded proteins (such as Alzheimer's, Parkinson's and prion disease), there is an early degeneration of presynaptic terminals prior to the loss of the neuronal somata. Identifying the mechanisms that govern synapse degeneration is of paramount importance, as cognitive decline is strongly correlated with loss of presynaptic terminals in these disorders. However, very little is known about the processes that link the presence of a misfolded protein to the degeneration of synapses. It has been suggested that the process follows a simple linear sequence in which terminals that become dysfunctional are targeted for death, but there is also evidence that high levels of activity can speed up degeneration. To dissect the role of activity in synapse degeneration, we infused the synaptic blocker botulinum neurotoxin A (BoNT/A) into the hippocampus of mice with prion disease and assessed synapse loss at the electron microscopy level. We found that injection of BoNT/A in naïve mice caused a significant enlargement of excitatory presynaptic terminals in the hippocampus, indicating transmission impairment. Long-lasting blockade of activity by BoNT/A caused only minimal synaptic pathology and no significant activation of microglia. In mice with prion disease infused with BoNT/A, rates of synaptic degeneration were indistinguishable from those observed in control diseased mice. We conclude that silencing synaptic activity neither prevents nor enhances the degree of synapse degeneration in prion disease. These results challenge the idea that dysfunction of synaptic terminals dictates their elimination during prion-induced neurodegeneration.

Redox stress and metal dys-homeostasis appear as hallmarks of early prion disease pathogenesis in mice.

PMID: 36170956 | Indexed in PubMed

Neurodegenerative diseases are associated with a multitude of dysfunctional cellular pathways. One major contributory factor is a redox stress challenge during the development of several protein misfolding conditions including Alzheimer's (AD), Parkinson's disease (PD), and less common conditions such as Creutzfeldt Jakob disease (CJD). CJD is caused by misfolding of the neuronal prion protein and is characterised by a neurotoxic unfolded protein response involving chronic endoplasmic reticulum stress, reduced protein translation and spongiosis leading subsequently to synaptic and neuronal loss. Here we have characterised prion disease in mice to assess redox stress components including nitroergic and oxidative markers associated with neuroinflammatory activation. Aberrant regulation of the homeostasis of several redox metals contributes to the overall cellular redox stress and we have identified altered levels of iron, copper, zinc, and manganese in the hippocampus of prion diseased mice. Our data show that early in disease, there is evidence for oxidative stress in conjunction with reduced antioxidant superoxide dismutase mRNA and protein expression. Moreover, expression of divalent metal transporter proteins was reduced for Atp7b, Atox1, Slc11a2, Slc39a14 at 6-7 weeks but increased for Slc39a14 and Mt1 at 10 weeks of disease. Our data present evidence for a strong pro-oxidant environment and altered redox metal homeostasis in early disease pathology which both may be contributory factors to aggravating this protein misfolding disease.

Aquaporin-4 mis-localization slows glymphatic clearance of α -synuclein and promotes α -synuclein pathology and aggregate propagation.

PMID: 39229234 | Indexed in PubMed

The appearance of misfolded and aggregated proteins is a pathological hallmark of numerous neurodegenerative diseases including Alzheimer's disease and Parkinson's disease. Sleep disruption is proposed to contribute to these pathological processes and is a common early feature among neurodegenerative disorders. Synucleinopathies are a subclass of neurodegenerative conditions defined by the presence of α -synuclein aggregates, which may not only enhance cell death, but also contribute to disease progression by seeding the formation of additional aggregates in neighboring cells. The mechanisms driving intercellular transmission of aggregates remains unclear. We propose that disruption of sleep-active glymphatic function, caused by loss of precise perivascular AQP4 localization, inhibits α -synuclein clearance and facilitates α -synuclein propagation and seeding. We examined human post-mortem frontal cortex and found that neocortical α -synuclein pathology was associated with AQP4 mis-localization throughout the gray matter. Using a transgenic mouse model lacking the adapter protein α -syntrophin, we observed that loss of perivascular AQP4 localization impairs the glymphatic clearance of α -synuclein from intersitial to cerebrospinal fluid. Using a mouse model of α -synuclein propagation, using pre-formed fibril injection, we observed that loss of perivascular AQP4 localization increased α -synuclein aggregates. Our results indicate α -synuclein clearance and propagation are mediated by glymphatic function and that AQP4 mis-localization observed in the presence of human synucleinopathy may contribute to the development and propagation of Lewy body pathology in conditions such as Lewy Body Dementia and Parkinson's disease.

α -Synuclein oligomers and clinical implications for Parkinson disease.

PMID: 23225525 | Indexed in PubMed

Protein aggregation within the central nervous system has been recognized as a defining feature of neurodegenerative diseases since the early 20th century. Since that time, there has been a growing list of neurodegenerative disorders, including Parkinson disease, which are characterized by inclusions of specific pathogenic proteins. This has led to the long-held dogma that these characteristic protein inclusions, which are composed of large insoluble fibrillar protein aggregates and visible by light microscopy, are responsible for cell death in these diseases. However, the correlation between protein inclusion formation and cytotoxicity is inconsistent, suggesting that another form of the pathogenic proteins may be contributing to neurodegeneration. There is emerging evidence implicating soluble oligomers, smaller protein aggregates not detectable by conventional microscopy, as potential culprits in the pathogenesis of neurodegenerative diseases. The protein α -synuclein is well recognized to contribute to the pathogenesis of Parkinson disease and is the major component of Lewy bodies and Lewy neurites. However, α -synuclein also forms oligomeric species, with certain conformations being toxic to cells. The mechanisms by which these α -synuclein oligomers cause cell death are being actively investigated, as they may provide new strategies for diagnosis and treatment of Parkinson disease and related disorders. Here we review the possible role of α -synuclein oligomers in cell death in Parkinson disease and discuss the potential clinical implications.